

PHISHING: HOW HACKERS STEAL YOUR DATA.



A COMPLETE GUIDE BY PARIS USMAN.

In 2017, Maersk, the world's largest global shipping company, fell victim to a cyberattack that resulted in a loss of over **\$300 million**.

This attack began with a simple phishing email—one of the most common entry points for cybercriminals. Below, we explore why phishing is so effective and how widespread it has become.

NotPetya, a **highly destructive malware**, was spread across computers through phishing emails containing malicious attachments.

The emails looked like normal business messages—invoices or fake job offers. When someone opened the file, the malware spread across the computer. Using a hacking tool, it moved freely across networks, spreading through Maersk's computers without needing passwords.

It locked employees’ computers and demanded **\$300 in Bitcoin** to ‘decrypt’ it. The hackers had no intention of decrypting the files: it was a trick to extort money without providing any actual solution.

Maersk was forced to shut down its systems, **leading to a loss of communication, logistics tracking, revenue and halting port operations.**

The result? An estimated **\$300 million** in losses including revenue losses and recovery costs.

PHISHING STATISTICS

While studies vary, **phishing emails and other forms of social engineering** account for over **90% of cyberattacks**.

The Federal Trade Commission report states consumers lost over \$10 billion to overall fraud in 2023, with emails becoming the number 1 contact method used by scammers to trick victims.	FTC, 2024.
More than 3.4 billion phishing emails are sent daily.	StationX.
Over 60% of business close down after a phishing attack.	The U.S National Cyber Alliance.

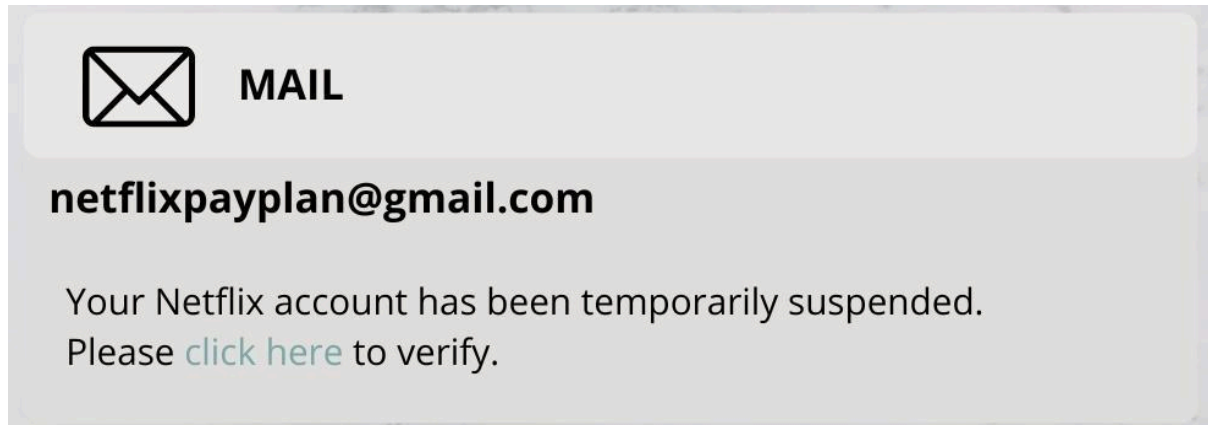
By the end of reading this guide, you’ll know exactly how to spot phishing emails and secure your data— even if you aren’t a tech expert.

WHAT IS PHISHING AND HOW DOES IT WORK?

Phishing messages are **fraudulent messages** designed to trick people into:

- revealing **sensitive information**—like passwords, credit card numbers or personal details.
- downloading **harmful malware**.

They often appear to come **from trusted sources** like banks, social media platforms, colleagues, or even IT support. A common example? An email notification saying your **Netflix account was suspended** with a link to ‘verify’ it.



The link could request for your login credentials or payment information, thus, stealing your data or money. Phishing emails could take place in different forms, some more subtle than others. Thus, the need to be able to spot them becomes critical.

THE PROCESS OF PHISHING.

Phishing is a step-by-step process. Here is how it works:

Step 1: Research.

Cyber attackers research their victims using publicly available information.

Example: A person's email address in their bio on their social media profile.

Step 2: Fake Identity.

They create fake email addresses that mimic legitimate communicators. These messages are often poorly written or edited.

Example: support@tecno.com -> Original email.

support@tecn0.com -> Phishing email

Step 3: Message composing.

They create messages which seem legitimate to their victims.

Example: A security alert or an invoice.

Step 4: **Apply pressure.**

They pressurize victims to act quickly and imbibe a feeling of urgency.

Example: An email from your co-worker requesting you make a 'due' payment immediately.

From: jessicamiller@company.com

To: annasantos@company.com

Subject: URGENT PAYMENT REQUIRED.

Hi, Anna

I hope this email finds you well. It's Jessica Miller from the Finance Team, and I'm reaching out regarding an urgent payment that requires your immediate attention.

Project Details:

Vendor: TechSolutions Ltd.

Invoice #: TS-2026-0742

Amount: \$5,000,000 (Five Million Dollars)

Due Date: 2026-07-09 (Today)

Payment Reference: "Writing Project"

Step 5: **Click or Reply.**

The victim clicks the link and downloads an attachment or replies with sensitive information.

Step 6: **The Attacker Strikes.**

Attackers steal credentials, implant malware or makes fraudulent transactions.

TYPES OF PHISHING MESSAGES

SPEAR PHISHING: This is a form of phishing where cyber attackers customize their messages for a specific **individual or organization**. They often use different modes of communications like calls, text messages and emails to create urgency.

'Whaling' is a type of spear fishing that targets high-profile individuals (eg CEO's, CFO's, Government officials etc)

Example (SPEAR FISHING): A work **email** from your company's 'IT department' **asking you** to 'update your password' due to a **security breach**.

POP UP PHISHING: This is a form of phishing where cyber attackers use pop-up windows that appear on legitimate websites to trick individuals into revealing sensitive information or downloading malware.

Example: A pop-up which claims you've won a prize and redirects you to another website, which requires your details so you can receive your prize.

SMS PHISHING (SMISHING): This is a form of phishing which uses text messages to trick people. The target is usually asked to click a link or contact a person whose details would be provided. They commonly pretend to be customer service representatives or financial services.

VOICE PHISHING (VISHING): In this form of phishing, cyber attackers make **automated phone calls**— often using **text to speech** or **voice automation**— to people in order to get individuals reveal personal information.

Example: A customer service representative from your bank calls to inform you that your account is on the verge of suspension and requires you to enter your details to fix it.

QR CODE PHISHING (QUISHING): In Quishing, cyber attackers use QR codes to trick individuals into giving up their personal data. When a person scans a **quish** code, they are automatically redirected to a malicious website, which was created to steal sensitive information like passwords, log in information and financial details.

Example: You scan a QR code of a restaurant poster to order food. You get **redirected to another website** which requires you to put in your **bank details** to **'make payment'** for your order.

MAN-IN-THE-MIDDLE (MITM): This type of phishing occurs when a hacker intercepts communication between two parties. The hacker uses tools like Wi-Fi spoofing and malicious hotspot.

Example: You connect to a free network labeled 'Airport Wi-Fi' and log into your bank or email on that network. The attacker can capture your credentials as you enter them, impersonate you to reset your passwords or steal money.

COMMON SIGNS OF A PHISHING ATTACK

URGENCY: Phishing attacks usually pressurize victims to act immediately with a threat of consequences. Example: "Your account will be closed in 24 hours, act now"

SUSPICIOUS SENDING ADDRESS: The attacker's mailing address looks similar to a legitimate one but not exact.

GENERIC GREETINGS: The greeting on the mail uses phrases like "Dear customer" instead of your actual name or username.

POOR GRAMMAR OR SPELLING: Phishing messages usually contain typos, bad phrasing or mistakes. However, not all of them are poorly written.

SUSPICIOUS LINKS OR ATTACHMENTS: When hovered over with a mouse, the link reveals a completely different and unusual page. Links from services like '[Bit.ly](#)' or 'TinyURL' that doesn't reveal its actual land pages.

OVERLY GENEROUS OFFERS: Messages claiming you've won a gift or prize from a giveaway you never entered. Scammers use this method to trick individuals to clicking malicious links or sharing financial data.

3 WAYS TO SPOT A PHISHING EMAIL IMMEDIATELY

1) Check the sender's address.

Always verify the senders email before clicking anything or sharing information.

- Look for mismatches: Phishing emails often use addresses that are similar to legitimate ones but with slight variations.
- Check the domain name: Ensure the domain matches the official website of the organization.

2) Inspect hyperlinks safely.

Don't click on a link in an email or message unless you've verified it's safe.

For Computers:

Hover your mouse cursor on it, (without clicking) to preview the URL. Legitimate links should start with 'https://' (the 's' stands for secure)

For Mobile devices:

Hold the link for a few seconds to see a preview of the URL. Use a URL expander tool if the link comes from services like '[bit.ly](#)' or '[tinyurl.com](#)'.

3) Watch for urgency or threats.

Phishing emails often create a sense of urgency to enable their victims act quickly. Watch for messages like 'Your account will be closed in 24 hours:[action required](#)'. Legitimate organizations won't ask you to provide sensitive information through emails.

4) Stay Informed.

Follow cybersecurity news to be updated on news tactics.

CYBERSECURITY 'PHISHING' QUIZ

Question 1: You receive an email from support@tecn0.com asking you to update your security settings. Based on the guide, what makes this address suspicious?

- A) The email domain ends in '.com' instead of '.org'.
- B) It contains a typo where the number zero (0) replaces the letter 'o'.
- C) Tech companies never use the word "support" in emails.
- D) The sender address is hidden from view.

Question 2: What is the specific name given to a targeted spear phishing attack that exclusively targets high-profile individuals like CEOs, CFOs, or government officials?

- A) Smishing
- B) Quishing
- C) Whaling
- D) Vishing.

Question 3: If an email contains a hidden link from a URL shortener service like bit.ly or tinyurl.com, what action does the guide recommend you take?

- A) Click the link on a mobile device instead of a computer.
- B) Reply to the email to ask the sender for the full web link.
- C) Copy and paste the link directly into your browser window.
- D) Use a URL expander tool to reveal the true destination link.

Question 4: In the 2017 Maersk cyberattack, how did the highly destructive NotPetya malware initially bypass computer security setups across networks?

- A) It guessed the administrative passwords of corporate executives.
- B) It exploited active network permissions and software vulnerabilities.
- C) It manually forced employees to type in their bank PIN numbers.
- D) It used automated phone calls to trick port staff out of data.

Question 5: How can you safely preview a destination web link on a mobile device without accidentally clicking it?

- A) Double-tap the link quickly.
- B) Turn your phone to landscape mode.
- C) Long-press the link for a few seconds.
- D) Take a screenshot of the message text.

QUIZ ANSWERS.

1. *B:* This is Step 2 in the phishing process. Hackers register lookalike domains with subtle typo tricks to fool you at a quick glance.
2. *C:* Whaling is a type of spear phishing designed to trick high-level executives who have deep access to company data and financial accounts.
3. *D:* URL shorteners mask the true web destination. Using an expander tool allows you to safely inspect the real domain string before navigating to it.
4. *B:* Once inside via a phishing attachment, NotPetya utilized network vulnerabilities to move freely across systems without needing password prompts.
5. *C:* Long-pressing a link on touchscreens opens a preview window displaying the full URL path, allowing you to check for threats safely.

GLOSSARY

- Social engineering is a tactic used by cybercriminals to trick people into revealing sensitive information or performing harmful actions.